

Note 13.4 Curvature and Normal Vectors of a Curve

Curvature and Normal Vectors of a Curve

Definition of Curvature

If $\mathbf{T}$ is the unit vector of a smooth curve, the **curvature** function of the curve is

$$\kappa = \left| \frac{d\mathbf{T}}{ds} \right|$$

When the curvature is large, $\mathbf{T}$ turns sharply as the particle passes through P .

It is related to the changed distance rather than the time. It contains the information of the length.

Formula for Calculating Curvature

If $\mathbf{r}(t)$ is a smooth curve, then the curvature is

$$\kappa = \left| \frac{d^2\mathbf{r}(s)}{ds^2} \right| = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right|$$

where $\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$ is the unit tangent vector.

Definition of Principal Unit Normal

At a point where $\kappa \neq 0$, the **principal unit normal** vector for a smooth curve in the plane is

$$\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds}$$

It contains the information of the direction. It can create a coordinate system with the position of the particle as the origin.

Formula for Calculating $\mathbf{N}$

If $\mathbf{r}(t)$ is a smooth curve, then the principal normal is

$$\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$$

where $\mathbf{T} = \mathbf{v}/|\mathbf{v}|$ is the unit tangent vector.

Circle of Curvature for Plane Curves

The **circle of curvature** or **osculating circle** at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

- is tangent to the curve at P . (has the same tangent line the curve has)
- has the intersection point P with the curve. (has the same tangent line the curve has)
- has the same curvature the curve has at P .

Exercises

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